



1.10

Rational Functions and Holes

Student Notes and Guided Practice

AP Precalculus - Unit 1

| ECHS Mathematics

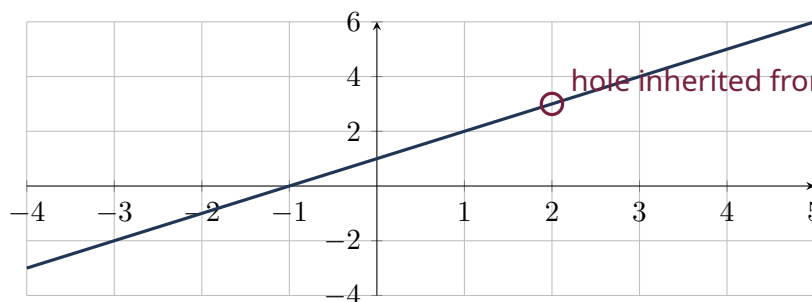
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Learning Targets

- Identify holes and determine their coordinates.
- Distinguish removable discontinuities from vertical asymptotes.
- Use equivalent formulas while preserving the original domain.

Essential Question

What information is lost when a rational expression is simplified without recording restrictions?



Concept Framework

Key Idea

A canceled factor records an excluded input in the original function.

Key Idea

The hole y-coordinate comes from the simplified formula evaluated at the excluded input.

Key Idea

Equivalent expressions need not define equal functions if their domains differ.

Key Idea

A removable discontinuity can be “filled” by defining the missing value appropriately.

Formula Map**Formula and Structure**

$$R(x) = \frac{(x-a)P(x)}{(x-a)Q(x)}, \quad x \neq a$$

Formula and Structure

$$\text{hole at } \left(a, \frac{P(a)}{Q(a)}\right) \text{ if } Q(a) \neq 0$$

Worked Examples**Worked Example 1**

Problem. Find the hole of $\frac{x^2-9}{x-3}$.

Solution. For $x \neq 3$, the function equals $x + 3$. The hole is at $(3, 6)$.

Worked Example 2

Problem. Define $F(2)$ to make $F(x) = \frac{x^2-4}{x-2}$ continuous.

Solution. The simplified expression is $x + 2$; set $F(2) = 4$.

Worked Example 3

Problem. Can a rational function have a hole and a vertical asymptote at the same x -value?

Solution. Not after full simplification: a factor either cancels completely (hole) or remains in the denominator (vertical asymptote). Partial cancellation can reduce multiplicity but a remaining factor still gives an asymptote, not a hole at that same input.

Multiple-Representation Connection**AP Reasoning**

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Find every hole and any remaining vertical asymptote of $\frac{x^2-16}{x-4}$.

2. Find every hole and any remaining vertical asymptote of $\frac{(x+2)(x-5)}{x+2}$.

3. Find every hole and any remaining vertical asymptote of $\frac{x^2+x-6}{x-2}$.

4. Find every hole and any remaining vertical asymptote of $\frac{(2x-1)(x+3)}{x+3}$.

5. Find every hole and any remaining vertical asymptote of $\frac{(x-1)^2(x+4)}{(x-1)(x+2)}$.

6. Define $F(3)$ to remove the discontinuity in $F(x) = \frac{x^2-9}{x-3}$.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.